SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency X → Y means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference	BL2 – Understand	5

8	If R(A,B,C) has FD: A→B and B→C, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	**5**
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	**5**
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	**5**

Code of Conduct:

I MONIKA.B (192571002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: MONIKA.B

RUBRICS FOR CALCULATION:

MCQ					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. Which of the following is true about 1NF?

(a) Allows multi-valued attributes

(b) Eliminates partial dependencies

(c) All attributes must be atomic

(d) Eliminates transitive dependencies

Answer: (c) All attributes must be atomic

2. A relation is in 2NF if it is in 1NF and:

(a) Has no transitive dependencies

(b) Has no partial dependencies on primary key

(c) Every attribute is a prime attribute

(d) Has no multi-valued dependencies

Answer: (b) Has no partial dependencies on primary key

3. Which normal form deals with multi-valued dependencies?

(a) 2NF

(b) 3NF

(c) BCNF

(d) 4NF

Answer: (d) 4NF

4. BCNF is a stricter version of:

(a) 1NF

(b) 2NF

(c) 3NF

(d) 4NF

Answer: (c) 3NF

5. Functional dependency X → Y means:

(a) Y determines X

(b) X determines Y

(c) X and Y are unrelated

(d) Both determine each other

Answer: (b) X determines Y

6. Which of the following anomaly is NOT associated with unnormalized relations?

(a) Insertion anomaly

(b) Deletion anomaly

(c) Update anomaly

(d) Retrieval anomaly

Answer: (d) Retrieval anomaly

7. A superkey is:

(a) A minimal set of attributes that uniquely identifies tuples

(b) Any set of attributes that uniquely identifies tuples

(c) An attribute that depends on a non-prime attribute

(d) A foreign key reference

Answer: (b) Any set of attributes that uniquely identifies tuples

8. If R(A,B,C) has FD: A→B and B→C, then C is:

(a) Partially dependent on A

(b) Fully dependent on A

(c) Transitively dependent on A

(d) Independently determined

Answer: (c) Transitively dependent on A

9. 5NF is also known as:

(a) Domain-Key Normal Form

(b) Boyce-Codd Normal Form

(c) Project-Join Normal Form

(d) Element Normal Form

Answer: (c) Project-Join Normal Form

10. Lossless-join decomposition ensures:

(a) No data is lost during decomposition

(b) All FDs are preserved

(c) The relation is in BCNF

(d) No null values exist

Answer: (a) No data is lost during decomposition